**THREE COLORS OF MADNESS**

***Technical Design Document***

**Internal working document. Not for external distribution. Companion to the Design Bible.**

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## Contents



# Part One — What This Document Is

**This is the Technical Design Document (TDD) for Three Colors of Madness. It is the companion to the Design Bible, not a replacement for any part of it.**

**The division between the two documents is simple. The bible owns the what and the why — thesis, Design Laws, narrative, the meaning of every mechanic. This document owns the how — engine architecture, data schemas, production sequencing, and the specific open questions a playtest still needs to settle before a mechanism the bible describes gets its final shape. Where a passage in the bible states a mechanism rather than just a rule, that is deliberate: the mechanism itself is what's enforcing a Design Law, and this document does not relitigate it. Where the two documents ever appear to disagree, the bible is correct and this document is wrong until revised — never the reverse.**

**This document is expected to change often and change fast. Unlike the bible, it has no standing rule against being provisional — an architecture decision made here can be wrong, discovered to be wrong, and replaced without that being a failure of the document. Its job is to stay honest about the current state of the build, not to be permanently settled.**

This revision grounds the document in a fourth pass over the same lineage: the same Godot branch, now carrying a further Codex pass on top of Claude Code's bug-fix content — three new content/correctness fixes, a genuine automated regression-test suite for two of them, and the branch's first playable Web export, which this revision verifies directly through a live browser session rather than by trusting the build's own self-report. Everything in Parts Two and Three below is a description of what the build actually does, verified by reading its source and commit history directly, or by playing the shipped Web build directly. Where something described in an earlier revision has since changed, been resolved, been superseded, or been corrected, that is called out explicitly rather than silently overwritten, so the document's own history of decisions and mistakes stays legible.

# Part Two — Current Build

## Engine and Target

Godot 4.7 (the Web export identifies as 4.7.2.stable), using the GL Compatibility renderer (not Forward+), targeting broad hardware reach over cutting-edge rendering fidelity — confirmed directly in the Web build's own console output, which reports WebGL 2.0 via the Compatibility backend. Window target 1440×900, canvas-items stretch mode. No import dependencies beyond the engine itself. Unchanged since v1.

## Scope of the Current Slice

The build still covers the opening movement of Chapter One ("No Exit Wound") — the Ophion estate grounds, Pickman Street, the precinct, the boardinghouse, Walter's room — and the service-passage encounter beneath the estate's kitchen wing described in this document's v2. This pass fixes three concrete correctness issues in content added by the prior pass, rather than adding new scenes.

- *Odell's answer can no longer be silently re-litigated.* `_odell_response()` now checks whether either of Odell's two recorded statements already exists before presenting the choice again; if one does, it shows an "Already answered" panel instead of the original two-button choice, so a player revisiting the captain after answering can't produce a second, contradictory statement. The prior build allowed the choice to be asked again indefinitely.
- *The woman outside Kessler's shop now actually leaves.* Her scene already stated, narratively, that she disappears after her warning; the code did not previously enforce that. A new `estate.gd`/`town.gd` method, `dismiss_old_woman()`, removes her interaction point and her figure once her scene is discovered, and is now called from every path that could otherwise re-show her — first discovery,

save/load, and travel back into town. This is verified by new assertions in `tests/town_flow.gd` (see Verification, below) rather than left to observation.
- ***The barman's "club talk" scene is retimed to match its own staging.*** **The scene was written as an evening bar conversation but is reached, in the current build, during a daytime portico encounter; the dialogue and its framing text ("BENEATH THE PORTICO" rather than "AT THE FAR END OF THE BAR") now describe the encounter that actually happens, and the question Walter asks shifted from present tense ("what the members talk about") to past tense ("what the members used to talk about") to match.**

**The build's own `docs/ARCHITECTURE.md` documents these three fixes plus two clarifications that were not code changes at all, just corrections to how the mechanism should be understood: the grounds crew (groundskeeper) observation added in the prior pass has no Perception gate and is available to any player regardless of accumulated evidence; and Perception, throughout the current build, gates *route information* (specifically, the wear-marks paragraph behind the tunnel's service screens) rather than *physical access* — nothing in the current build blocks movement or a location outright behind a Perception check. Both are confirmed directly against source in this pass: `target("crew", ...)` in `estate.gd` is registered unconditionally alongside every other observation point, with no surrounding `if state.perception() >= N` guard, and the only `perception() >= 4` check in `scripts/chapters/chapter_one.gd` gates one paragraph of descriptive text, not a door, a route, or an interaction.**

**Not yet implemented, unchanged from v2: combat, the forced-spill mechanic, the glass-shattering ontological break, the fatal-comprehension ending, Chapters Two and Three, gamepad support, full key rebinding, a full encumbrance/leveling system, and Ekon's later consumption of Walter's record. The Part Seven difficulty screen is deferred. None of this pass's work touches the chapter's actual climax; it corrects three specific defects in the town/estate content that already existed.**

# Controls and Accessibility

**Unchanged from v2.**

# Saves

**Unchanged in mechanism from v2. The old woman's departure is itself now saved state — `dismiss_old_woman()` mutates the same `estate.points` dictionary and is re-applied on load rather than the removal being purely a runtime/session effect — so a player who saves after her scene and reloads does not see her reappear. No schema change to `case_state.gd` was needed.**

## Verification

This pass ships with genuine automated regression coverage for both of its behavioral fixes, not just its content. `tests/opening_flow.gd` now saves and reloads mid-test specifically to assert Odell's locked answer survives a save/load round trip, and separately asserts that none of the dialogue buttons shown afterward begin with the original "Say nothing" option — the test fails if the lock can be bypassed by reopening the conversation. `tests/town_flow.gd` asserts the old woman's interaction point is gone immediately after her scene, gone again after an explicit save/load cycle, gone again after traveling away from and back into town, and that attempting to re-trigger her observation directly does not open a dialogue page. These are meaningfully specific assertions about state persistence, not smoke-test traversal. A `tests/capture_views.gd` addition also stages a dedicated camera position and lighting setup for a future "observer" QA screenshot — this pass adds the capture rig but does not include a rendered output image in this delivery; see Part Six.

# Part Three — Architecture As Built

This section describes what the current codebase actually does, as read directly from the project and its commit history, plus what a live session against the shipped Web export actually showed. Where something differs from an earlier revision's Architecture As Built, that's noted explicitly; everything else in Part Three (autoloads, entry point/shared-module split, case-state shape, environment, the Observer color-preservation shader, dialogue architecture) is unchanged from v5 and not repeated here.

## Web Export and Live Browser Verification — new

This is the branch's first commit to include an actual Web/WebAssembly export (`build/web/`), and this revision is the first to verify a build of this project by playing it directly, in a live browser session, rather than by reading source alone. The verification found:

- The title, accessibility-and-controls, chapter framing, and opening intertitle sequence all loaded and read exactly as the source predicts — including the "Eight people are dead at the Ophion estate. The town is prepared to account for six" framing and the prologue's "Those who came home disagreed about what they had seen" line, both consistent with Design Law 4.
- Movement, camera control, examine, and the case file all functioned correctly once reached in gameplay. The rose garden crime scene was reached and explored directly: the "Examine the six men" prompt, the named-victim breakdown

(Judge Wexford, Dr. Fenn, Corliss the district attorney, Pruitt, Kessler, plus one unidentified sixth man), and Judge Wexford's specific examine text — "A wound above the bridge of the nose. No powder scorching. Walter turns the head. There is no exit wound." — all matched source exactly.

- Movement reliability was inconsistent earlier in this verification pass, in a way initially suspected to need a code-level fix (an input-latch on the movement controller, to guard against a held key being polled as released within a single physics frame). Direct comparison against this pass's actual commit history finds no such change: nothing in `scripts/chapters/chapter_one.gd` or any other script touches `_physics_process`, `Input.is_action_pressed`, or the movement/input path at all in this pass's commits. The far more likely explanation, consistent with when movement started working reliably during testing, is that keeping the browser window/pane in the foreground was what mattered, not any change to the game's own input handling. This document is not crediting a movement fix to the game code, because the evidence doesn't support one having been made.

This is a genuine milestone for verification going forward — future passes can be checked by playing the actual Web export rather than relying solely on source reading and the build's own self-report, the same standard already applied to git history in this document.

# Part Four — Architecture Decisions Needed

Items resolved since v5 are marked as such rather than deleted, so the decision history stays visible. Items unchanged from v5 are restated briefly for continuity; see v5's fuller reasoning if needed.

Odell's answer could be silently re-litigated — resolved. See Part Two. `_odell_response()` now locks against either recorded statement.

The old woman outside Kessler's shop did not actually leave — resolved. See Part Two. `dismiss_old_woman()` now enforces the departure the scene already narrated, across discovery, save/load, and travel.

The barman's club-talk scene was staged for the wrong time of day — resolved. See Part Two.

Shader color-preservation scope — unchanged from v5; still open. The active shader preserves color through a screen-wide red-dominance test, supported by a controlled art palette, with no object-specific mask. A capture rig for a dedicated "observer" QA

**screenshot was added this pass (`tests/capture_views.gd`), but no rendered output from it shipped in this delivery — see Part Six. True object-specific isolation remains a separate rendering decision if later assets or lighting make palette-based selection unreliable.**

**`_town_observation()`'s hardcoded return-to-witness argument — unchanged from v5; still open.**

**NPCs have no general scheduling or staging trigger system — unchanged in substance, with one narrow exception now in the codebase. `dismiss_old_woman()` is, as far as this document has found, the first piece of conditional NPC-presence logic anywhere in the build: it removes one specific NPC's interaction point and figure once a specific piece of state (her scene being discovered) becomes true, and re-applies that removal on load and on travel. This is not a general scheduling system — it's a single hand-written method for a single NPC's departure, with no reusable concept of "this figure exists only once X is true" that a future NPC could opt into — but it is a real, working precedent for exactly the kind of state-gated NPC presence this document has been asking for since v5, and worth using as a starting reference point rather than building the eventual general system from nothing. The gardener and every other always-present figure in `estate.gd`'s `_ready()` are otherwise unchanged: still placed unconditionally, with no gating logic at all.**

**Perception gates route information, not physical access — clarified, not new. The build's own `docs/ARCHITECTURE.md` states this explicitly as a correction to how the mechanism should be understood, and this document confirms it directly against source: the only `perception() >= 4` check in `chapter_one.gd` gates a single descriptive paragraph, not a door, a route, or an interaction point. Nothing about how the current build actually behaves changed; only the risk of describing it as an access gate is now closed off.**

**Two coexisting case-tracking systems — still open, unchanged from v2. No change this pass.**

**Procedural geometry: blockout strategy or production pipeline? — still open, unchanged from v2. No change this pass.**

# Part Five — Open Production Questions

**Unchanged from v5 in substance; nothing in this pass bears directly on Chapter Two's, Chapter Three's, or the meta layer's open questions.**

# Part Six — Next Steps

**In rough priority order: resolve the corkboard/link-mechanic decision, still the single most consequential open item. Decide the shape of a general NPC scheduling/staging system — `dismiss_old_woman()` is now a working single-NPC precedent worth generalizing rather than a theoretical gap. Capture and review the actual "observer" QA screenshot the new capture rig in `tests/capture_views.gd` was built to produce; it does not exist in this delivery, so the shader's real-world legibility against the current palette is still unconfirmed by any image, only by reading the formula. Decide whether the palette-discipline convention for color isolation needs to become an actual object/material mask before more Observer content is added. Generalize `_town_observation()`'s return-to-menu logic before a third revisitable town menu gets added. Decide the procedural-art-pipeline question before committing more chapters to code-generated geometry. Extend the new live-Web-export verification practice going forward — this pass showed it catches real things (the input-reliability question) that source reading alone could not have settled either way. Once a Chapter Two slice exists, this document gets a matching Architecture As Built entry for it, held to the shared/chapter-specific template v2 established.**